

OOPs STUDY MATERIAL

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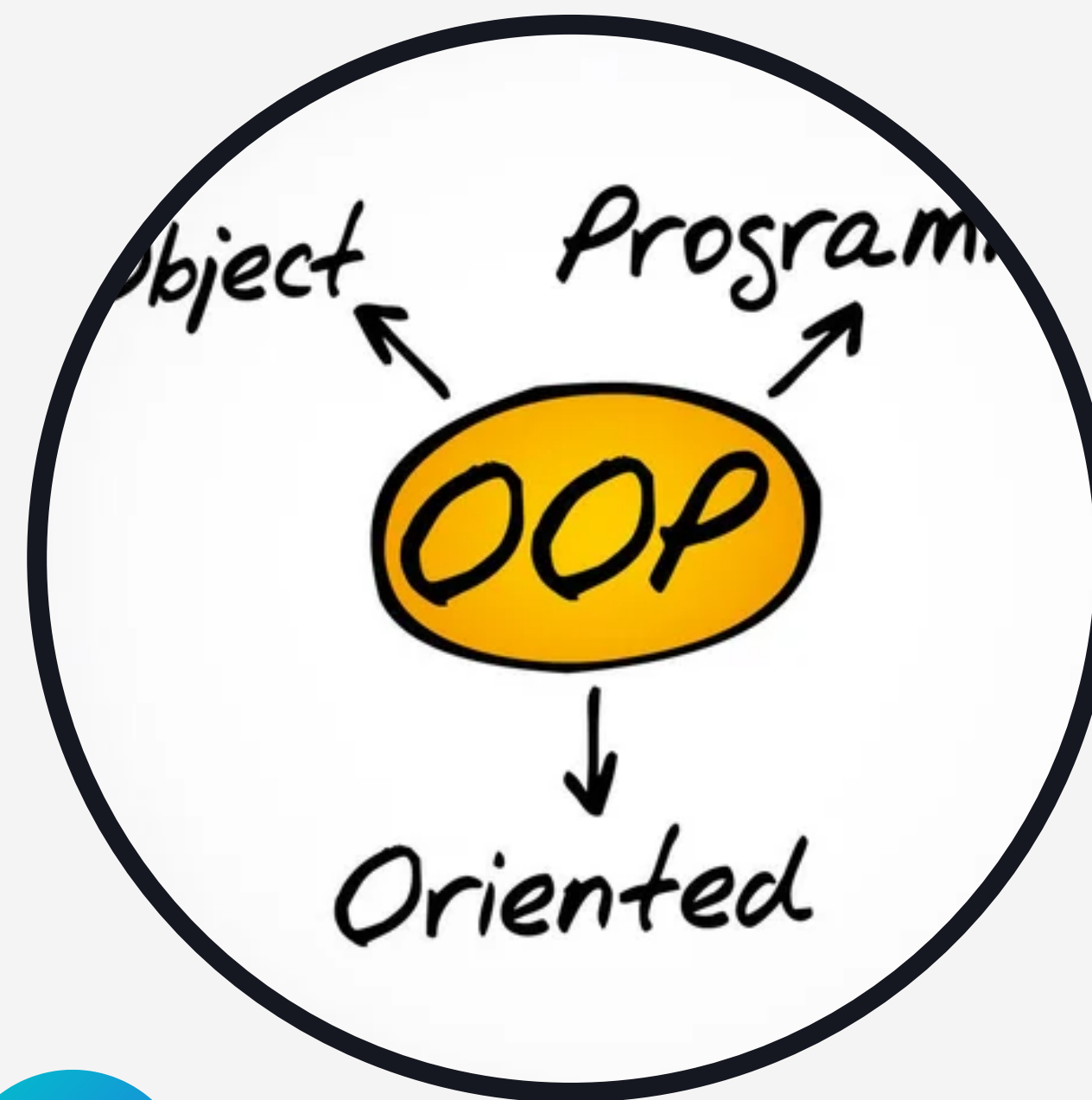
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OOPS – OBJECT ORIENTED PROGRAMMING :

OOPs stands for object oriented programming principle.

There are four pillars of OOPs :

1. Encapsulation
2. Polymorphism
3. Inheritance
4. Data abstraction



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Class is the collection of Data Members and Members Function.

Data Members : These are the attributes and properties of a class.

Members Function : These are the functions which are performed by the object of a class.

Object : Object is the instance of the CLASS . i.e., it is a real-life entity that represent a class.



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Data Members & Member Functions

Class Student

```
{  
    int roll, class, age;  
    string name, grade;  
    float percentage;  
}
```

Data Members

```
    void study()  
    {  
        =====  
    }  
    void play()  
    {  
        =====  
    }  
}
```

Member Functions



ENCAPSULATION:

The wrapping up or bundling up of Data Members and Member Functions into a single unit is called Encapsulation.

The best advantages of Encapsulation is security, because we can restrict the flow of our data by using access modifiers i.e. public, private, protected, and default.

What is static?

If something is static, then it is not compulsory to call it with an object, we can call it directly by using the class name.

Real-life example of Encapsulation :

Capsule: Just like in capsule various medicines are capsulated inside the shell protects from external environment. Similarly, access modifiers inside a class prevents unauthorized access of data.

POLYMORPHISM :

Poly means “Many” and morph means “Forms”;
The ability to acquire multiple forms is called Polymorphism.

Real-life example of polymorphism:

The real-life example of polymorphism can be “Umbrella” which can be used in rainy season to protect us from rain and it can be also used in summer season to protect us from the heat waves of sun.

Polymorphism are of two types :

1. Compile-time Polymorphism
2. Run-time Polymorphism

COMPILE TIME POLYMORPHISM:

Compile-time Polymorphism are of three types:

1. Function Overloading: When there are various functions with the same name but different arguments in the same class then it is known as the function overloading.
2. Constructor Overloading: When there are various constructor with the same name but different arguments then it is called as constructor overloading.
3. Operator Overloading: When a single operator can perform multiple operations, then it is called operator overloading.



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RUN TIME POLYMORPHISM :

Run-time Polymorphism includes function overriding:

Function Overriding:

When there are two functions with the same name and same arguments but present in different classes, one is present in parent class and other one is redefined in a child class, so this process of redefining is called Function Overriding.

INHERITANCE :

The property to inherit the properties of other classes is called inheritance.

The best advantage of inheritance is **Code Reusability**.

Types of Inheritance :

1. Single Inheritance
2. Multilevel Inheritance
3. Multiple Inheritance
4. Hierarchical Inheritance
5. Hybrid Inheritance

TYPES OF INHERITANCE :

1. Single Inheritance: In this Inheritance we have only one child class and one parent class.
 - Parent class is known as the base class.
 - Child class is known as the derived class.
2. Multi-level Inheritance: Multi-level Inheritance is just like Grandfather, Father and Son relationship, A class can extend the property of another class and that another class extend the property of some other class, this is called as Multi-level Inheritance.
3. Multiple Inheritance: When there is a single child class which extends the property of multiple parent classes then it is called as multiple inheritance.
4. Hierarchical Inheritance: In the hierarchical Inheritance, there is only one parent class but multiple child classes.
5. Hybrid Inheritance: It is the combination of any two types of inheritance.

DATA ABSTRACTION:

Showing only abstract idea and hiding the implementation is called Data Abstraction .

There are two ways to achieve data abstraction.

1. Interfaces: It is a way to achieve 100% data abstraction, because all the method inside the interface are only declared not even a single method is defined.
2. Abstract Class: It is a way to achieve the partial data abstraction because inside an abstract class some method are declared and some method are defined.

NOTE : We cannot create object of interfaces and abstract class .

ABOUT SAI CRT:

OOPS NOTES

Sai Campus Recruitment Training is the best institute in Central India for Campus Placements located in Indrapuri, Bhopal. CRT stands for Campus Recruitment Training batch. This is a program where we teach students everything from basics to advance required to get placed in world's top MNC's like Amazon, TCS, Infosys, SAP, Capgemini, Accenture, Cognizant etc. After joining this batch you don't have to worry about anything, We cover every single thing required for your smooth selection. 2000+ students are already placed in their dream MNC's....., YES! you are next!

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ABOUT GAURAV TIWARI SIR:



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Mr. Gaurav Tiwari, a multi-faceted individual with a diverse range of talents and achievements. Mr. Tiwari is widely recognized as a prominent programming trainer, aptitude and logical reasoning instructor, and a proficient communicator. As a famous educator and emerging entrepreneur, he has left an indelible mark in the world of education and business.

Beyond his expertise in training, Mr. Gaurav Tiwari's passion for chess has led him to achieve international recognition as a rated chess player. He has represented himself in numerous national and international chess events, showcasing his strategic acumen and dedication to the game.

As a testament to his innovative abilities, Mr. Tiwari led the team "Ingenious," which emerged victorious in the prestigious Smart India Hackathon 2017, a platform where young minds collaborate to solve real-world challenges through technology.

His outstanding achievements have not gone unnoticed, as he was honored by the Governor of Madhya Pradesh State for securing the highest number of job offers from top multinational companies. This accolade exemplifies his exceptional skills and dedication in the corporate realm.



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ABOUT GAURAV TIWARI SIR:



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Apart from his educational pursuits and entrepreneurial ventures, Mr. Gaurav Tiwari shares his experiences and knowledge with the world as a daily vlogger on his YouTube channel "Gauraverse." Through this platform, he has inspired and motivated thousands of viewers.

In a groundbreaking move, Mr. Tiwari established "Cryptovilla Cafe," India's first cryptocurrency-based cafe, demonstrating his forward-thinking approach to business and technology.

Moreover, his passion for coding and love for teaching have driven him to mentor over 10,000 students, making a lasting impact on the lives and careers of aspiring coders.

Mr. Gaurav Tiwari's remarkable journey and versatile talents make him an extraordinary individual, leaving an indelible mark in the domains of education, entrepreneurship, chess, and technology.



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